

Chapter Two: Mobile Digital Telecommunications

2.1. Introduction

Mobile Digital Telecommunication is the provision of telephone services to phones, which may move around freely, rather than stay fixed in one location. Mobile phones connect to a terrestrial cellular network of base stations (cell sites), whereas satellite phones connect to orbiting satellites. Both networks are interconnected to the public switched telephone network (PSTN) to allow any phone in the world to be dialed. Mobile phones send and receive radio signals with any number of cell site base stations fitted with microwave antennas. These sites are usually mounted on a tower, pole or building, located throughout populated areas, then connected to a cabled communication network and switching system. The phones have a low-power transceiver that transmits voice and data to the nearest cell sites, normally not more than 8 to 13 km (approximately 5 to 8 miles) away. In areas of low coverage, a cellular repeater may be used, which uses a long distance high-gain dish antenna or yagi antenna to communicate with a cell tower far outside of normal range, and a repeater to rebroadcast on a small short-range local antenna that allows any cell phone within a few meters to function properly.

In the early 1980s, most mobile telephone systems were analog rather than digital. One challenge facing analog systems was the inability to handle the growing capacity needs in a cost-efficient manner. As a result, digital technology was welcomed. The advantages of digital systems over analog systems include ease of signaling, lower levels of interference, integration of transmission and switching, and increased ability to meet capacity demands.

2.1.1. How it works?

When the mobile phone or data device is turned on, it registers with the mobile telephone exchange, or switch, with its unique identifiers, and can then be alerted by the mobile switch when there is an incoming telephone call. The handset constantly listens for the strongest signal being received from the surrounding base stations, and is able to switch seamlessly between sites. As the user moves around the network, the “handoffs” are performed to allow the device to switch sites without interrupting the call.

Cell sites have relatively low-power (often only one or two watts) radio transmitters which broadcast their presence and relay communications between the mobile handsets and the switch. The switch in turn connects the call to another subscriber of the same wireless service provider or to the public telephone network, which includes the networks of other wireless carriers. Many of these sites are camouflaged to blend with existing environments, particularly in scenic areas.

The dialogue between the handset and the cell site is a stream of digital data that includes digitized audio (except for the first generation analog networks). The technology that achieves this depends on the system which the mobile phone operator has adopted. The technologies are

grouped by generation. The first-generation systems started in 1979 with Japan, are all analog and include AMPS and NMT. Second-generation systems, started in 1991 in Finland, are all digital and include GSM, CDMA and TDMA.

2.2. Global System for Mobile Communication (GSM)

Throughout the evolution of cellular telecommunications, various systems have been developed without the benefit of standardized specifications. This presented many problems directly related to compatibility, especially with the development of digital radio technology. The GSM standard is intended to address these problems.

GSM is a globally accepted standard for digital cellular communication. It is the name of standardization group established in 1982 to create a common European mobile telephone standard that would formulate specifications for a pan-European mobile cellular radio system operating at 900 MHz. GSM was first deployed in Finland in December 1991. As of 2014, it has become the global standard for mobile communications – with over 90% market share, supporting more than one billion mobile subscribers in more than 210 countries worldwide.

GSM provides recommendations, not requirements. The GSM specification define functions and interface requirements in details but do not address the hardware. The reason for this is to limit designers as little as possible but still to make it possible for the operators to buy equipment from different suppliers. GSM provides basic to advanced voice and data services including **roaming** service. **Roaming** is the ability to use your GSM phone number in another GSM network.

2.2.1. Why GSM?

Listed below are the features of GSM that account for its popularity and wide acceptance.

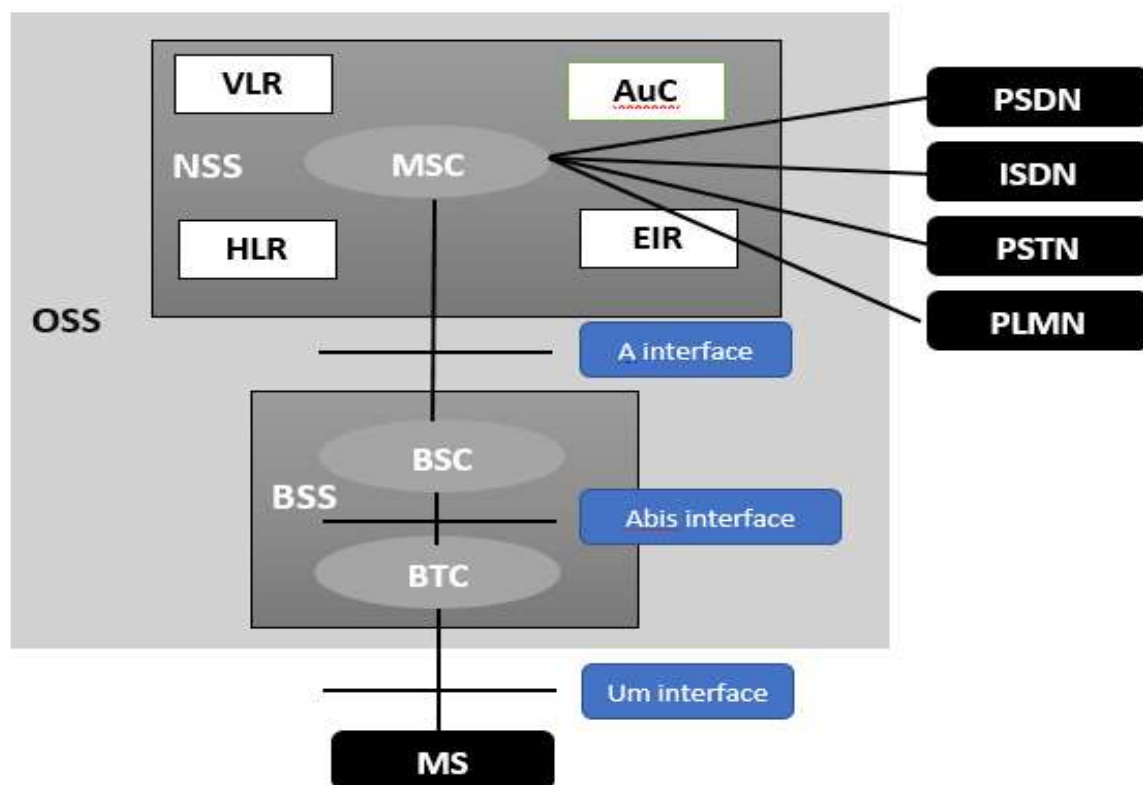
- Improved spectrum efficiency
- International roaming
- Low-cost mobile sets and base stations BSs
- High-quality speech
- Compatible with Integrated Services Digital Network (ISDN) and other telephone company services
- Support for new service

2.2.2. GSM – Architecture

A GSM network comprises of many functional units. These functions and interfaces are explained in this chapter. The GSM network can be broadly divided into:

- Mobile Station (MS)
- Base Station Subsystem (BSS)
- Network Switching Subsystem (NSS)
- Operation Support Subsystem (OSS)

Given below is a simple pictorial view of the GSM architecture.



- **Mobile Station (MS):** A GSM subscriber requires a terminal called MS to connect to the network using the radio interface (Um).
- **Base Station Subsystem (BSS):** The BSS performs radio-related functions. It consists of BTSs (Base Transceiver Stations) and BSCs (Base Station Controllers).
- **Network Switching Subsystem (NSS):** is responsible for call control, service control and **subscriber mobility management functions.**
- **Operation Support Subsystem (OSS):**

The additional components of the GSM architecture comprise of databases and messaging systems functions:

- **Base Transceiver Station (BTS):** handles the radio interface to the MS. It consists of radio equipment (transceivers and antennas) required to service each cell in the network.
- **Base Station Controller (BSC):** provides the control functions and physical links between the MSC and the BTS. A numbers of BSCs are served by one MSC while several BTSs can be controlled by one BSC.
- **Home Location Register (HLR):** HLR is a database used for storage and management of subscriptions. The HLR is considered the most important database, as it stores *permanent* data about subscribers, including a subscriber's service profile, location information, and activity status. When an individual buys a subscription from one of the PCS operators, he/she is registered in the HLR of that operator.
- **Mobile services switching center (MSC):** MSC performs telephony switching functions of the system. It controls calls to/from other telephone and data systems. It also performs functions like toll ticketing, network interfacing, common channel signaling, and others.
- **Visitor Location Register (VLR):** VLR is a database that contains temporary information about subscribers that it needed by the MSC in order to service visiting subscribers. The VLR is always integrated with the MSC. When a mobile station roams (travels) into a new MSC area, the VLR connected to the MSC will request data about the mobile station from the HLR. Later, if the mobile station makes a call, the VLR will have the information needed for call setup without having to interrogate the HLR each time.
- **Equipment Identity Register (EIR):** is a database that contains information about identity of mobile equipment that prevents calls from stolen, unauthorized, or defective mobile stations. AUC and EIR are implemented as stand-alone nodes or as a combined AUC/EIR node.
- **Authentication Center (AuC):** a unit called AuC provides authentication and encryption parameters that verify the user's identity and ensure the confidentiality of each call. AUC protects network operators from different types of fraud found in today's cellular world.
- **SMS Serving Center (SMSSC):**
- **Gateway MSC (GMSC):** is an MSC that serves as a gateway node to external networks, such as ISDN or wireline networks.
- **Chargeback Center (CBC):**
- **Transcoder and Adaptation Unit (TRAU):**

The MS and the BSS communicate across the *Um interface*. The *Um interface* is also known as the *air interface* or the *radio link*. The BSS communicates with the Network Service Switching (NSS) center across the *A interface*.

Subscriber Identity Module (SIM)

One of the key features of GSM is the Subscriber Identity Module, commonly known as a **SIM card**. The SIM is a detachable smart card containing the user's subscription information and phone book. This allows the user to retain his or her information after switching handsets. Alternatively, the user can change operators while retaining the handset simply by changing the SIM. Some operators will block this by allowing the phone to use only a single SIM, or only a SIM issued by them; this practice is known as ***SIM locking***.

2.2.3. GSM Network Areas

In a GSM network, the following areas are defined:

- **Cell:** Cell is the basic service area; one BTS covers one cell. Each cell is given a Cell Global Identity (CGI), a number that uniquely identifies the cell.
- **Location Area:** A group of cells form a Location Area (LA). This is the area that is paged when a subscriber gets an incoming call. Each LA is assigned a Location Area Identity (LAI). Each LA is served by one or more BSCs.
- **MSC/VLR Service Area:** The area covered by one MSC is called the MSC/VLR service area.
- **PLMN:** The area covered by one network operator is called the Public Land Mobile Network (PLMN). A PLMN can contain one or more MSCs.

2G networks developed as a replacement for first generation (1G) analog cellular networks, and the GSM standard originally described a digital, circuit-switched network optimized for full duplex voice telephony. This expanded over time to include data communications, first by circuit-switched transport, then by packet data transport via GPRS (General Packet Radio Services) and EDGE (Enhanced Data rates for GSM Evolution, or EGPRS).

Subsequently, the 3GPP developed third-generation (3G) UMTS standards, followed by fourth-generation (4G) LTE Advanced standards, which do not form part of the ETSI GSM standard.

2.2.4. GSM - Addresses and Identifiers

GSM treats the users and the equipment in different ways. Phone numbers, subscribers, and equipment identifiers are some of the known ones. There are many other identifiers that have been well-defined, which are required for the subscriber's mobility management and for addressing the remaining network elements. Vital addresses and identifiers that are used in GSM are addressed below.

1. International Mobile Station Equipment Identity IMEI

The IMEI looks more like a serial number which distinctively identifies a **mobile station** internationally. This is allocated by the equipment manufacturer and registered by the network operator, who stores it in the Entrepreneurs-in-Residence EIR. By means of IMEI, one recognizes obsolete, stolen, or non-functional equipment. Following are the parts of IMEI:

- **Type Approval Code TAC:** 6 decimal places, centrally assigned.
- **Final Assembly Code FAC:** 6 decimal places, assigned by the manufacturer.
- **Serial Number SNR:** 6 decimal places, assigned by the manufacturer.
- **Spare SP:** 1 decimal place.

Thus, $IMEI = TAC + FAC + SNR + SP$. It uniquely characterizes a mobile station and gives clues about the manufacturer and the date of manufacturing.

2. International Mobile Subscriber Identity IMSI

Every **registered user** has an original International Mobile Subscriber Identity (IMSI) with a valid IMEI stored in their Subscriber Identity Module (SIM).

IMSI comprises of the following parts:

- **Mobile Country Code (MCC):** 3 decimal places, internationally standardized.
- **Mobile Network Code (MNC):** 2 decimal places, for unique identification of mobile network within the country.
- **Mobile Subscriber Identification Number (MSIN):** Maximum 10 decimal places, identification number of the subscriber in the home mobile network.

3. Mobile Subscriber ISDN Number (MSISDN)

The authentic telephone number of mobile station is Mobile Subscriber ISDN Number (MSISDN).

Based on the SIM, a mobile station can have many MSISDNs, as each subscriber is assigned with a separate MSISDN to their SIM respectively.

Listed below is the structure followed by MSISDN categories, as they are defined based on international ISDN number plan:

- **Country Code (CC):** Up to 3 decimal places.
- **National Destination Code (NDC):** Typically 2-3 decimal places.
- **Subscriber Number (SN):** Maximum 10 decimal places.

2.2.5. GSM Operations

Once a Mobile Station initiates a call, a series of events takes place. Analyzing these events can give an insight into the operation of the GSM system.

i. Mobile Phone to Public Switched Telephone Network (PSTN)

When a mobile subscriber makes a call to a PSTN telephone subscriber, the following sequence of events takes place:

1. The MSC/VLR receives the message of a call request.
2. The MSC/VLR checks if the mobile station is authorized to access the network. If so, the mobile station is activated. The service will be denied otherwise.
3. MSC/VLR analyzes the number and initiates a call setup with the PSTN.
4. MSC/VLR asks the corresponding BSC to allocate a traffic channel (a radio channel and a timeslot).
5. The BSC allocates the traffic channel and passes the information to the mobile station.
6. The called party answers the call and the conversation takes place.
7. The mobile station keeps on taking measurements of the radio channels in the present cell and the neighboring cells and passes the information to the BSC. The BSC decides if a handover is required. If so, a new traffic channel is allocated to the mobile station and the handover takes place. If handover is not required, the mobile station continues to transmit in the same frequency.

ii. PSTN to Mobile Phone

When a PSTN subscriber calls a mobile station, the following sequence of events takes place:

1. The Gateway MSC receives the call and queries the HLR for the information needed to route the call to the serving MSC/VLR.
2. The GMSC routes the call to the MSC/VLR.
3. The MSC checks the VLR for the location area of the MS.
4. The MSC contacts the MS via the BSC through a broadcast message, that is, through a paging request.
5. The MS responds to the page request.
6. BSC allocates traffic channel and sends message to MS to tune to the channel. The MS generates ringing signal and, after the subscriber answers, speech connection established.
7. Handover, if required, takes place, as discussed in the earlier case.

2.2.6. GSM - User Services

GSM offers much more than just voice telephony. Contact your local GSM network operator to the specific services that you can avail. GSM offers three basic types of services:

- Telephony services or teleservices
- Data services or bearer services

- Supplementary services

i. Teleservices

The abilities of a Bearer Service are used by a Teleservice to transport data. These services are further transited in the following ways:

- **Voice Calls:** The most basic Teleservice supported by GSM is telephony. This includes full-rate speech at 13kbps and emergency calls, where the nearest emergency-service provider is notified by dialing three digits.
- **Videotext and Facsimile:** Another group of teleservices includes Videotext access, Teletex transmission etc.
- **Short Text Messages:** Short Messaging Service (SMS) service is a text messaging service that allows sending and receiving text messages on your GSM mobile phone. In addition to simple text messages, other text data including news, sports, financial, language, and location-based data can also be transmitted.

ii. Bearer Services

Data services or Bearer Services are used through a GSM phone. To receive and send data is the essential building block leading to widespread mobile Internet access and mobile data transfer. GSM currently has a data transfer rate of 9.6k. New developments that will push up data transfer rates for GSM users are High Speed Circuit Switched Data (HSCSD) and General Packet Radio Service (GPRS) are now available.

iii. Supplementary Services

Supplementary services are additional services that are provided in addition to teleservices and bearer services. These services include caller identification, call forwarding, call waiting, multiparty conversations, and barring of outgoing international calls, among others. A brief description of supplementary services is given here:

- **Conferencing:** It allows a mobile subscriber to establish a multiparty conversation, i.e. a simultaneous conversation between three or more subscribers to setup a conference call. This service is only applicable to normal telephony.
- **Call Waiting:** This service notifies a mobile subscriber of an incoming call during a conversation. The subscriber can answer, reject, or ignore the incoming call.
- **Call Hold:** This service allows a subscriber to put an incoming call on hold and resume after a while. The call hold service is applicable to normal telephony.

- **Call Forwarding:** It is used to divert calls from original recipient to another number. It is set up by subscriber himself. It can be used by the subscriber to divert calls from Mobile Station when the subscriber is not available, and so to ensure that calls are not lost.
- **Call Barring:** Call Barring is useful to restrict certain types of outgoing calls such as ISD or stop incoming calls from undesired numbers. Call barring is a flexible service that enables the subscriber to conditionally bar calls.

2.2.7. GSM Billing

GSM service providers are doing billing based on the services they are providing to their customers. All the parameters are simple enough to charge a customer for the provided services. This section provides an overview of the frequently used billing techniques and parameters applied to charge a GSM subscriber.

i. Telephony Service

These services can be charged on per call basis. The call initiator has to pay the charges, and the incoming calls are nowadays free. A customer can be charged based on different parameters like:

- International call or long distance call
- Local call
- Call made during peak hours
- Call made during night time
- Discounted call during weekends
- Call per minute or per second
- Many more other criteria can be designed by a service provider to charge their customers.

ii. SMS Service

Most of the service providers charge their customers' SMS services based on the number of text messages sent. There are other prime SMS services available where service providers charge more than normal SMS charge. These services are being availed in collaboration of Television Networks or Radio Networks to demand SMS from the audiences. Most of the time, the charges are paid by the SMS sender but for some services like stocks and share prices, mobile banking facilities, and leisure booking services, etc. the recipient of the SMS has to pay for the service.

iii. GPRS Services

Using GPRS service, you can browse, play games on the Internet, and download movies. So a service provider will charge you based on the data uploaded as well as data downloaded on your mobile phone. These charges will be based on per Kilo Byte data downloaded/uploaded.

iv. Supplementary Services

Most of the supplementary services are being provided based on monthly rental or absolutely free. For example, *call waiting*, *call forwarding*, *calling number identification*, and *call on hold* are available at zero cost.

- **Call barring** is a service, which service providers use just to recover their dues, etc., otherwise this service is not being used by any subscriber.
- **Call conferencing** service is a form of simple telephone call where customers charged for multiple calls made at a time. No service provider charges extra charge for this service.
- **Closed User Group (CUG)** is very popular and is mainly being used to give special discounts to the users if they are making calls to a particular defined group of subscribers.
- **Advice of Charge (AoC)** can be charged based on number of queries made by subscriber.

2.3. General Packet Radio System (GPRS)

What is GPRS?

- GPRS is a packet oriented data service for IP and X.25 over GSM networks. It enables packet-switched services on the already existing GSM network infrastructure. Deploying new channel coding schemes and timeslot bundling, GPRS is capable of providing single user throughput rates of up to 160 kbps (in theory). It requires a major hardware upgrade in the GSM network and it requires new mobile stations. It is a step toward 3G networks (2.5G)

2.3.1. Who owns GPRS?

GPRS specifications are written by the European Telecommunications Standard Institute (ETSI), that is the European counterpart of the American National Standard Institute ANSI.

2.3.2. Key Features

Following three key features describe wireless packet data

- **The always online feature:** Removes the dial-up process, making applications only one click away.
- **An upgrade to existing systems:** Operators do not have to replace their equipment; rather, GPRS is added on top of the existing infrastructure.
- **An integral part of future 3G systems:** GPRS is the packet data core network for 3G systems EDGE and WCDMA.

2.3.3. Goals of GPRS

GPRS is the first step toward an end-to-end wireless infrastructure and has the following goals:

- Open architecture
- Consistent IP services
- Same infrastructure for different air interfaces
- Integrated telephony and Internet infrastructure
- Leverage industry investment in IP
- Service innovation independent of infrastructure

2.3.4. Benefits of GPRS

- **Higher Data Rate:** GPRS benefits the users in many ways, one of which is higher data rates in turn of shorter access times. In the typical GSM mobile, setup alone is a lengthy process, and equally rates for data permission are restrained to 9.6 kbps. The session establishment time offered while GPRS is in practice is lower than one second and ISDN-line data rates are up to many 10 kbps.
- **Easy Billing:** GPRS packet transmission offers a more user-friendly billing than that of circuit switched services. In circuit switched services, billing is based on the duration of the connection. This is unsuitable for applications with bursty traffic. The user must pay for the entire airtime, even for the idle periods when no packet has been sent (e.g., when the user reads a Web page).

In contrast to this, with packet switched services, billing can be based on the amount of transmitted data. The advantage for the user is that he or she can be "online" over a long period of time but will be billed based on the transmitted data volume only.

2.3.5. GPRS – Characteristics and Applications

GPRS has opened a wide range of unique services to the mobile wireless subscriber. Some of the characteristics that have opened a market full of enhanced value services to the users. Below are some of the characteristics:

- **Mobility:** The ability to maintain constant voice and data communications on the move.
- **Immediacy:** Allows subscribers to obtain connectivity when needed, regardless of location and without a lengthy login session.
- **Localization** - Allows subscribers to obtain information relevant to their current location.

Using the above three characteristics, varied possible applications are being developed for the mobile subscribers.

- **Communications:** E-mail, fax, unified messaging, and intranet/internet access, etc.
- **Value-added services:** Information services, games, etc.

- **E-commerce:** Retail, ticket purchasing, banking and financial trading, etc.
- **Location-based applications:** Navigation, traffic conditions, airline/rail schedules, location finder, etc.
- **Vertical applications:** Freight delivery, fleet management, and sales-force automation.
- **Advertising:** Advertising may be location sensitive. For example, a user entering a mall can receive advertisements specific to the stores in that mall.

2.4. Advanced Mobile Telecommunication Standards: EDGE, UMTS

➤ Enhanced Data for Global Evolution (EDGE)

EDGE (also known as Enhanced GPRS or EGPRS) is a data system used on top of GSM networks. It provides nearly three times faster speeds than the outdated GPRS system. The theoretical maximum speed is 473 kbps for 8 timeslots but it is typically limited to 135 kbps in order to conserve spectrum resources. Both phone and network must support EDGE, otherwise the phone will revert automatically to GPRS. EDGE meets the requirements for a 3G network but is usually classified as 2.75G.

➤ Universal Mobile Telecommunications System (UMTS)

The Universal Mobile Telecommunications System (UMTS) is a third generation mobile cellular system for networks based on the GSM standard. Developed and maintained by the 3GPP (3rd Generation Partnership Project), UMTS is a component of the International Telecommunications Union IMT-2000 standard set and compares with the CDMA2000 standard set for networks based on the competing cdmaOne technology. UMTS uses wideband code division multiple access (W-CDMA) radio access technology to offer greater spectral efficiency and bandwidth to mobile network operators.

UMTS specifies a complete network system, which includes the radio access network (UMTS Terrestrial Radio Access Network, or UTRAN), the core network (Mobile Application Part, or MAP) and the authentication of users via SIM (subscriber identity module) cards.

The technology described in UMTS is sometimes also referred to as Freedom of Mobile Multimedia Access (FOMA) or 3GSM.

Unlike EDGE (IMT Single-Carrier, based on GSM) and CDMA2000 (IMT Multi-Carrier), UMTS requires new base stations and new frequency allocations.